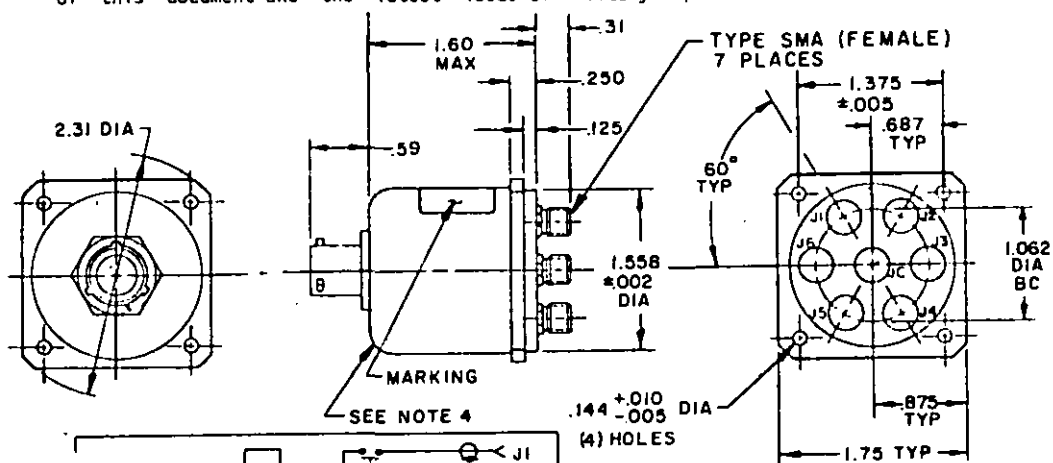
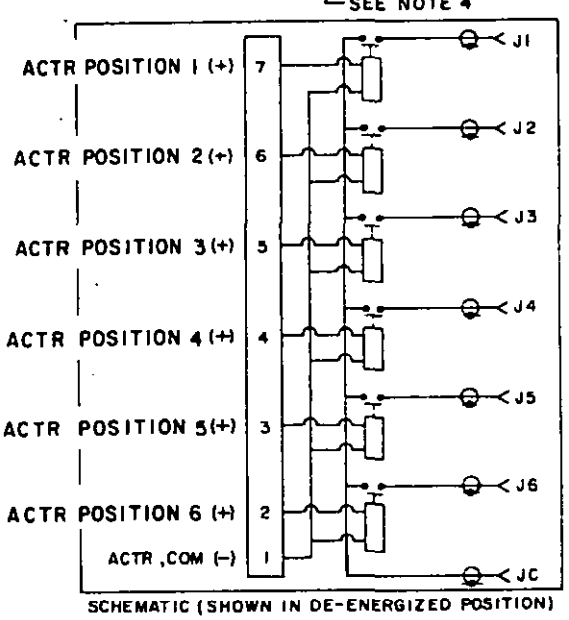
MILITARY SPECIFICATION SHEET

SWITCHES (COAXIAL), RADIO FREQUENCY TRANSMISSION LINE (ELECTRICALLY OPERATED) CLASS 5, 1P6T

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch described herein shall consist of this document and the latest issue of Military Specification MIL-S-3928.





SCHEMATIC (SHOWN IN DE-ENERGIZED POSITION)

INCHES	MM
.002	.05
.005	.13
.010	.25
.125	3.18
.144	3.66
.25	6.4
.31	7.9
.59	15.0
.687	17.45
.875	22.22
1.062	26.97
1.375	34.92
1.558	39.57
1.60	40.6
1.75	44.4
2.31	58.7

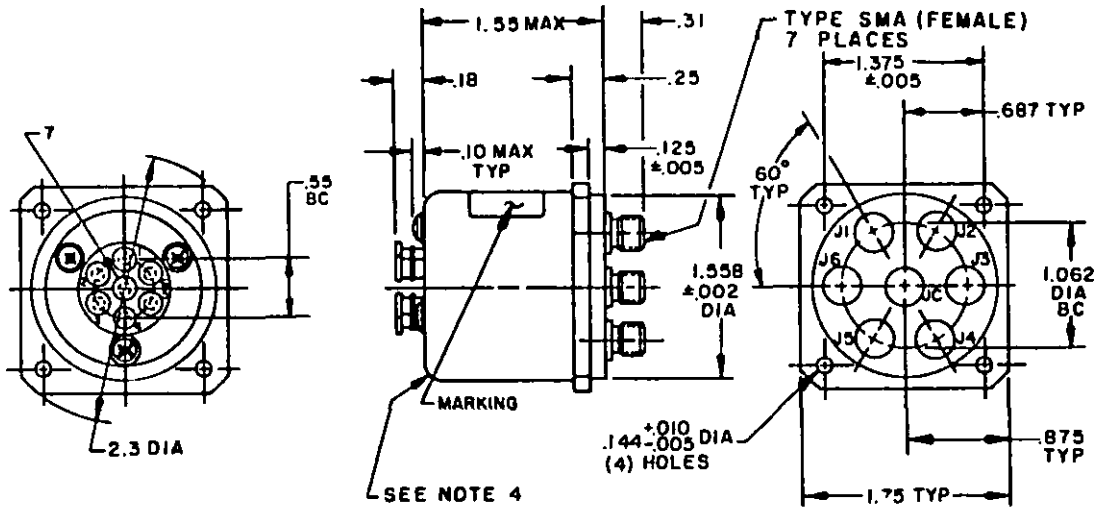
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
4. Round corners of case may be squared.

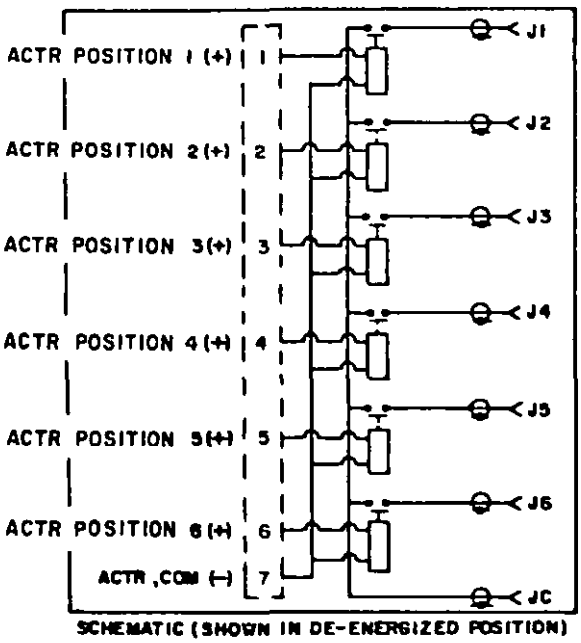
Ⓐ denotes changes

FIGURE 1. Switch configuration and schematic, part number M3928/18-01.

FSC 5985

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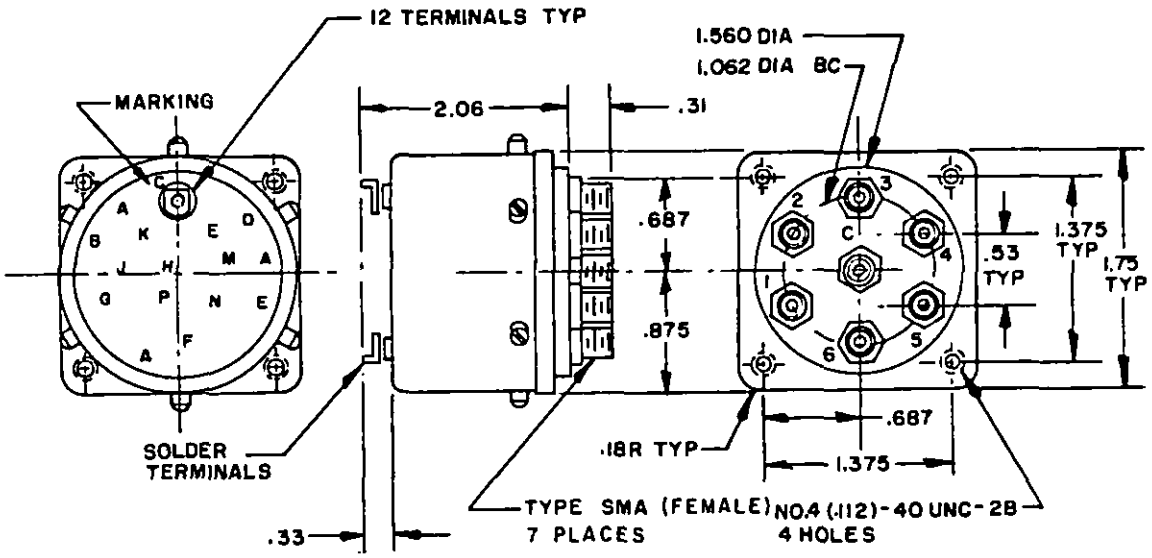
SCHEMATIC (SHOWN IN DE-ENERGIZED POSITION)

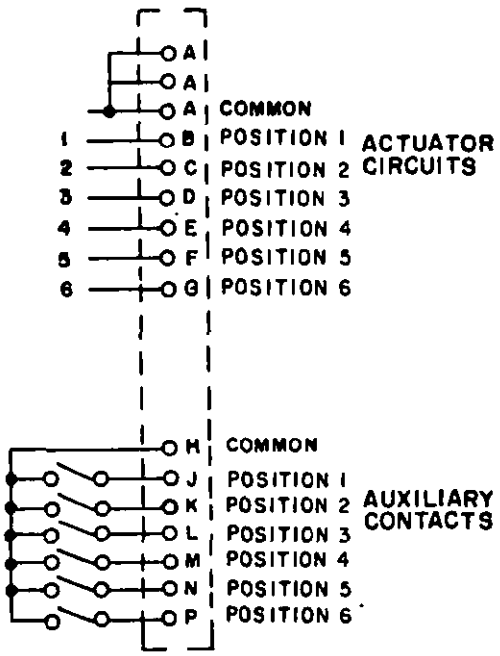
INCHES	MM
.002	.05
.005	.13
.010	.25
.10	2.5
.125	3.18
.144	3.66
.18	4.6
.25	6.4
.31	7.9
.55	14.0
.687	17.45
.875	22.22
1.062	26.97
1.375	34.92
1.55	39.4
1.558	39.57
1.75	44.4
2.3	58.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 2. Switch configuration and schematic, part number M3928/18-02.

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INCHES	MM
.112	2.84
.31	7.9
.33	8.4
.687	17.45
.875	22.22
1.062	26.97
1.375	34.92
1.560	39.62
1.75	44.4
2.06	52.3

SCHEMATIC

Ⓐ NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. Round corners of case may be squared.
5. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ±.02 (.5 mm) for two place decimals.

FIGURE 3. Outline and schematic drawing for P/N M3928/18-03.

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Ⓐ TABLE I. Electrical and performance characteristics.

Part no. M3928/19-	Figure no.	Housing	Frequency range dc to GHz	VSWR	Insertion loss (dB)	Isolation (dB)	Switch time (ms)	Position indicating circuit and rating	Life cycles (x 1000)	Failsafe or latching	Oper. current (amps) 1/	Holding current (amps) 1/	Nominal oper. voltage	Pickup voltage (less than)	Dropout voltage (less than)	Power and indicator connector	Weight (oz)
				max	max	min	max				max	max					max
01	1	I	18	1.5:1	0.5	60	20	None	1000	-	.17	.17	28 Vdc	20 Vdc	16 Vdc	RTK07-8-7P (Deutsch) or equal	6
02	2	I	18	1.5:1	0.5	60	20	None	1000	-	.17	.17	28 Vdc	20 Vdc	16 Vdc	Solder terminals	6
03	3	I	-18 @ 2-7 @ 2-3	1.5:1 1.25:1	0.5 0.3	2/ 60	50	None	1000	-	.22	.22	28Vdc	24Vdc	18Vdc 3/	Solder terminals	4.5

1/ At 28 Vdc and 20° C.
2/ Isolation between connected ports and open ports shall be 6 dB minimum.
3/ Switch must return to de-energized position at no less than 2 volts.

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REQUIREMENTS:

Dimensions and configuration: See figures 1 through 3.

RF connectors: Female connector (7 places) shall meet the requirements of MIL-C-39012 and shall mate with SMA type male connectors per MIL-C-39012/55.

Electrical and performance characteristics: See table I.

Ⓐ Auxiliary contacts: 2 amps @ 28 Vdc.

RF power handling capability (average): 25 watts (min) except for dash number 03 which is 30 watts.

Nominal impedance: 50 ohms.

Termination: Open.

Ⓐ Transient interference (RFI): Required for P/N M3928/18-03.

Altitude: 60,000 feet for P/N M3928/18-03.

Vibration: Method I.

Ⓐ Operating temperature: -55°C to +85°C except for dash number 03 which is -55°C to +115°C.

Part number: M3928/18- (dash number from table I).

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI,
Navy -
Air Force - 11, 17, 99
DLA - ES

User activities:
Army -
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5985-0878-3)

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